

# Probabilistic Machine Learning

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## Lecture 04: Independence, Naive Bayes and Graphical Models

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# Module 1

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## Bayes as a Classifier

**Guiding Question:** *The data tells us how attributes behave given the class. We need the reverse. What does the conversion cost?*

# Thinking With Uncertainty

Bayes' theorem lets us update a **hypothesis**  $H$  in the light of **data**  $D$ :

$$P(H \mid D) = \frac{P(H) \cdot P(D \mid H)}{P(D)}$$

This is the whole apparatus of evidence-based decision making in one line: a belief, a measurement of how well the belief explains what we saw, and a normalisation.

# Example: Do I Have the Flu?

You have a headache and a cough. From the literature:

## Likelihood

90% of people with the flu  
show headache and coughing

$$P(D \mid H) = 0.9$$

## Prior

5% of people currently have  
the flu

$$P(H) = 0.05$$

## Evidence

headache and coughing occur  
in 20% of people

$$P(D) = 0.2$$

**How likely is it that you have the flu?**

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How likely is it that you have the flu?

$$P(H \mid D) = \frac{0.05 \cdot 0.9}{0.2} = 0.225$$

# Why So Low?

## Few people have the flu

The prior  $P(H) = 0.05$  caps how far the evidence can carry us.

## The symptoms are common

$P(D) = 0.2$  means many people have these symptoms *without* the flu.

A symptom is informative to the degree that it is **rare in general** but **common under the hypothesis**. A likelihood alone tells you nothing.

# Takeaway

Bayesian decision making weighs **prior × likelihood** against how ordinary the evidence is. Deciding under uncertainty means keeping all three numbers in view, not just the one the study reports.

# Two Directions of the Same Connection

## Observed connection

$$A_1, \dots, A_p \mid B$$

*Given the class, how do the attributes behave?* — the standard situation, and the one data can answer directly.

## Reversed connection

$$B \mid A_1, \dots, A_p$$

*Given the attributes, which class?* — the diagnosis situation, and the one we actually need.

$$P(B \mid A_1, \dots, A_p) = \frac{P(B) \cdot P(A_1, \dots, A_p \mid B)}{P(A_1, \dots, A_p)}$$

# What the Classifier Needs

## Class priors

$$P(\text{class} = y)$$

How common is each class, before seeing any attribute?

## Attribute likelihoods

$$P(\text{attribute} = v \mid \text{class} = y)$$

How does each class distribute its attribute values?

Given both, the posterior  $P(\text{class} \mid \text{attributes})$  follows from Bayes — and the prediction is the class with the highest posterior.

Estimating  $P(A_1, \dots, A_p \mid B)$  requires the **joint** distribution of all attributes within each class: exponentially many parameters, and never enough data.

# Module 2

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## Naive Bayes

**Guiding Question:** *Its central assumption is almost always false, and it classifies well anyway. How can both be true?*

# Generative and Discriminative Classifiers

## Generative

Model the **joint** distribution  $P(A, B)$  — how the data was produced — then derive  $P(B | A)$ .

*Naive Bayes.*

## Discriminative

Model  $P(B | A)$  **directly** — only the boundary between classes.

*Logistic regression.*

# The Naive Assumption

Assume the attributes are **conditionally independent given the class**:

$$P(A_1, \dots, A_p \mid B) = \prod_{j=1}^p P(A_j \mid B)$$

Exponentially many parameters collapse to  $p$  per class — one distribution per attribute.

The assumption is almost always false. Naive Bayes nonetheless classifies well, because ranking the classes correctly is easier than estimating their probabilities correctly.

# The Classifier

$$P(B_i \mid A_1, \dots, A_p) = \frac{P(B_i) \cdot \prod_{j=1}^p P(A_j \mid B_i)}{\sum_{i=1}^k P(B_i) \cdot \prod_{j=1}^p P(A_j \mid B_i)}$$

## Score each class

prior × product of attribute likelihoods

## Normalise

divide by the total over all classes — the theorem of total probability

## Predict

the class with the largest posterior

For prediction alone the denominator can be skipped — it is the same for every class. It is needed only when the probability itself is reported.

# Example: Enjoy Sport?

| Outlook  | Temperature | Humidity | Wind   | EnjoySport |
|----------|-------------|----------|--------|------------|
| sunny    | hot         | high     | weak   | no         |
| sunny    | hot         | high     | strong | no         |
| overcast | hot         | high     | weak   | yes        |
| rain     | mild        | high     | weak   | yes        |
| rain     | cool        | normal   | weak   | yes        |
| rain     | cool        | normal   | strong | no         |

Estimate the priors and the per-attribute likelihoods by **counting**, then score a new day.

# The Zero-Frequency Problem

If an attribute value never occurs with a class in the training data, its estimated likelihood is **zero** — and one zero factor drives the whole product to zero, whatever the other attributes say.

**Laplace smoothing** adds a pseudo-count to every combination, so that unseen does not mean impossible.

**One missing count**

vetoed every other piece of evidence

Naive Bayes buys tractability with a **conditional independence assumption it knows to be false** — and works anyway, because classification needs the right argmax, not the right probability.

# Module 3

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## Independence, Properly

***Guiding Question:** Naive Bayes assumed independence to survive. What does the assumption actually claim — and when does conditioning create dependence rather than remove it?*

# Conditional Independence

**Definition.** Events  $A$  and  $B$  are **conditionally independent given  $C$**  (written  $A \perp\!\!\!\perp B \mid C$ ) iff

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C)$$

equivalently

$$P(A \mid B \cap C) = P(A \mid C)$$

Reading: **once  $C$  is known,  $B$  carries no further information about  $A$ .**

# Dependent, but Independent Given C

Let  $P(A) = \frac{8}{16}$ ,  $P(B) = \frac{3}{16}$ ,  $P(C) = \frac{4}{16}$  on a 16-cell sample space.

## Unconditionally dependent

$$P(A \cap B) = \frac{1}{16}$$

$$P(A) \cdot P(B) = \frac{3}{32}$$

$$\frac{1}{16} \neq \frac{3}{32}, \text{ so } A \not\perp B$$

## Conditionally independent given $C$

$$P(A \cap B \mid C) = \frac{1}{4}$$

$$P(A \mid C) \cdot P(B \mid C) = \frac{2}{4} \cdot \frac{2}{4} = \frac{1}{4}$$

$$\text{so } A \perp\!\!\!\perp B \mid C$$

Also directly:  $P(A \mid B \cap C) = \frac{1}{2} = P(A \mid C)$ .

# Why This Matters

## Explaining away

A common cause  $C$  can make two effects look related.

## Model size

Conditional independence lets a joint distribution factorise into small pieces.

## Later in the course

Naive Bayes assumes it.  
Graphical models are built from it.

Conditional independence neither implies nor is implied by ordinary independence. Both directions can fail.

# Takeaway

Conditioning can **create** independence as easily as it destroys it. "Given what I already know, does this extra fact still tell me anything?" is the question that makes large probabilistic models tractable.

# Module 4

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## From Assumption to Model

**Guiding Question:** *Naive Bayes assumes everything is independent given the class. What if we could state exactly which independences hold?*

# The Problem: Joint Distributions Explode

Consider a joint distribution  $p(x_1, \dots, x_n)$  over  $n$  **binary** random variables. How many configurations are there?

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Consider a joint distribution  $p(x_1, \dots, x_n)$  over  $n$  **binary** random variables. How many configurations are there?

$2^n$  states,  $2^n - 1$  free parameters

## Specification

For large  $n$ , writing down all values is impossible.

## Inference

A single marginal  $p(x_1)$  requires summing over  $2^{n-1}$  states.

## Chain rule?

It rewrites the joint exactly — but does not reduce the number of values.

# What a Graphical Model Is

**Definition (Graphical model).** A graphical model depicts the dependence and independence relations of a distribution using a graph  $G$  as its underlying structure.

## Modelling

Describe the variables of a domain and their interactions — how  $p(x_1, \dots, x_n)$  **factorises**.

## Inference

Answer queries such as  $p(x_1 \mid x_2 = a, \dots)$  efficiently over that factorisation.

Usually one class of model is used for modelling and another for inference the latter is often compiled from the former.

# How a Graph Can Help

Probabilistic inference is summing and multiplying. Let every variable be a node, and read edges as interaction:

## No edge

$$f(a, b) = g(a) \cdot h(b)$$

The variables are independent — the factor splits.

## Edge

$$f(a, b) \neq g(a) \cdot h(b)$$

The variables are dependent — the factor does not split.

Edge **direction** can encode further structure. The principle extends to any number of variables.

Graphical models have limited expressive power: not every conditional independence statement can be drawn, and different graphs can represent the same distribution.

# Example: Why Is the Grass Wet?

Tracey leaves her house and sees that her grass is wet. **Was it rain, or did she leave the sprinkler on?** She notices her neighbour Jack's grass is also wet.

*R*

raining (1) or not (0)

*S*

Tracey's sprinkler on  
(1) or off (0)

*J*

Jack's grass wet (1)  
or dry (0)

*T*

Tracey's grass wet  
(1) or dry (0)

# Building the Model

The full joint over four binary variables needs  $2^4 - 1 = 15$  probabilities. The chain rule rewrites it but does not shrink it:

$$p(T, J, R, S) = p(T \mid J, R, S)p(J \mid R, S)p(R \mid S)p(S)$$

Now bring in **background knowledge** — assumptions of conditional independence:

Tracey's grass does not depend on Jack's

$$p(T \mid J, R, S) = p(T \mid R, S)$$

Jack's grass does not depend on Tracey's sprinkler

$$p(J \mid R, S) = p(J \mid R)$$

Rain is not influenced by the sprinkler

$$p(R \mid S) = p(R)$$

$$p(T, J, R, S) = p(T \mid R, S)p(J \mid R)p(R)p(S)$$

Only  $4 + 2 + 1 + 1 = 8$  probabilities remain — and each one is a question a person can actually answer.

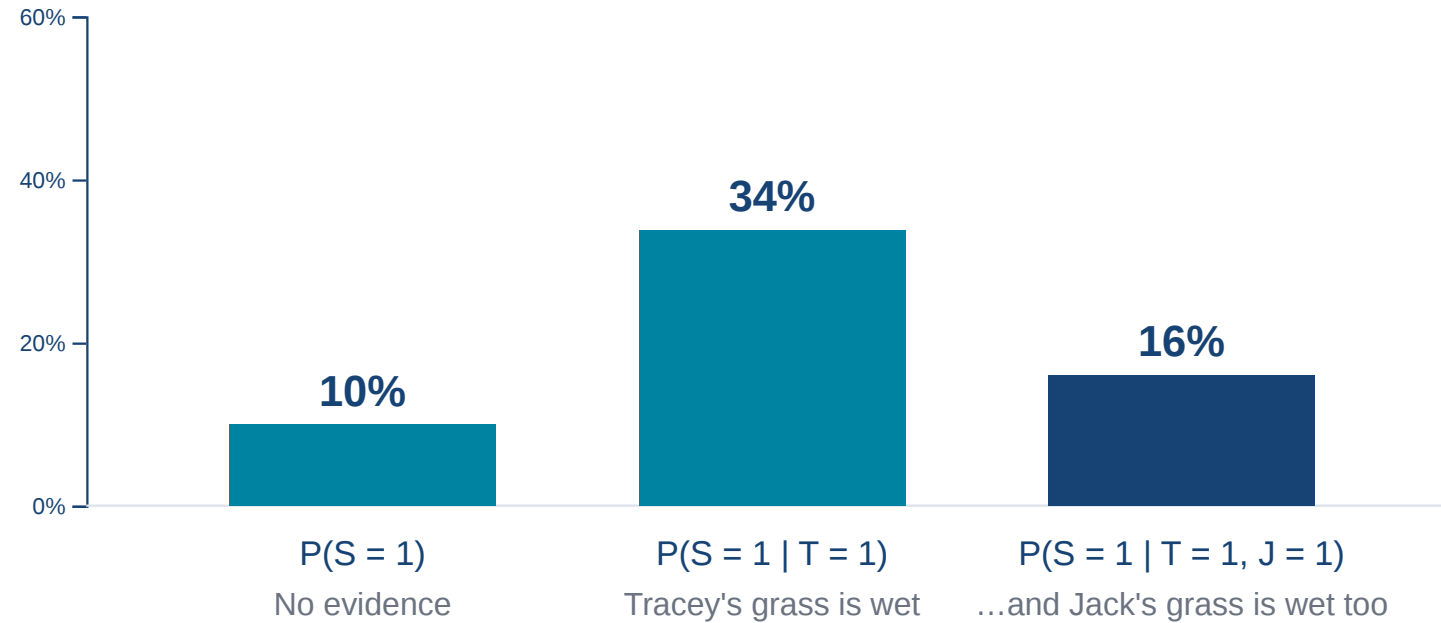
# The Probability Tables

| Factor                       | $p(\cdot)$ | $1 - p(\cdot)$ |
|------------------------------|------------|----------------|
| $p(R = 1)$                   | 0.2        | 0.8            |
| $p(S = 1)$                   | 0.1        | 0.9            |
| $p(J = 1 \mid R = 0)$        | 0.2        | 0.8            |
| $p(J = 1 \mid R = 1)$        | 1.0        | 0.0            |
| $p(T = 1 \mid R = 1, S = 0)$ | 1.0        | 0.0            |
| $p(T = 1 \mid R = 1, S = 1)$ | 1.0        | 0.0            |
| $p(T = 1 \mid R = 0, S = 1)$ | 0.9        | 0.1            |
| $p(T = 1 \mid R = 0, S = 0)$ | 0.0        | 1.0            |

Missing edges in the graph are exactly the assumptions above the tables supply the numbers.

# Inference: What Does the Evidence Say?

Rain explains the wetness — the sprinkler falls back towards its prior



# Explaining Away

$$p(S = 1 \mid T = 1) = \frac{\sum_{J,R} p(T=1 \mid R, S=1)p(J \mid R)p(R)p(S=1)}{\sum_{J,R,S} p(T=1 \mid R, S)p(J \mid R)p(R)p(S)}$$

Two independent causes ( $R$  and  $S$ ) become **dependent once their common effect is observed**. Learning that rain occurred lowers the probability of the sprinkler, even though rain and sprinkler are unconditionally independent.

# Takeaway

A graphical model turns **background knowledge into a factorisation**: missing edges are conditional independence assumptions, and each one removes parameters. The graph is a picture of how the joint splits — nothing more, and nothing less.